

visuals cdc 26

```
library(tidyverse)
library(broom)
library(knitr)

complaints_by_zip <- readRDS("complaints_by_zip.rds")

model_white <- readRDS("model_white.rds")
model_black <- readRDS("model_black.rds")
model_asian <- readRDS("model_asian.rds")
model_hispanic <- readRDS("model_hispanic.rds")
model_other_race <- readRDS("model_other_race.rds")
model_two_or_more <- readRDS("model_two_or_more.rds")

# Create income in $10,000 units
complaints_by_zip <- complaints_by_zip %>%
  mutate(
    income_10k = median_household_income / 10000
  )

# Keep only observations with valid values for analysis
analysis_zcta <- complaints_by_zip %>%
  filter(
    !is.na(complaints_per_1000),
    is.finite(complaints_per_1000),
    complaints_per_1000 >= 0,
    !is.na(population),
    population > 0
  )
```

```
race_results <- bind_rows(
  tidy(model_white, conf.int = TRUE) %>%
    filter(term == "white_percent") %>%
    mutate(race = "White"),

  tidy(model_black, conf.int = TRUE) %>%
    filter(term == "black_percent") %>%
    mutate(race = "Black"),

  tidy(model_asian, conf.int = TRUE) %>%
    filter(term == "asian_percent") %>%
    mutate(race = "Asian"),

  tidy(model_hispanic, conf.int = TRUE) %>%
    filter(term == "hispanic_percent") %>%
    mutate(race = "Hispanic"),
```

```

tidy(model_other_race, conf.int = TRUE) %>%
  filter(term == "other_race_percent") %>%
  mutate(race = "Other race"),

tidy(model_two_or_more, conf.int = TRUE) %>%
  filter(term == "two_or_more_races_percent") %>%
  mutate(race = "Two or more races")
)

```

```

library(ggplot2)
library(dplyr)
library(scales)

use_custom_font <- TRUE

if (use_custom_font) {
  library(showtext)
  library(sysfonts)
  font_add_google("Inter", "inter")
  showtext_auto()
  base_font <- "inter"
} else {
  base_font <- "sans"
}

race_results <- race_results %>%
  mutate(
    significant = !(conf.low <= 0 & conf.high >= 0),
    direction = case_when(
      estimate > 0 ~ "Higher complaint rate",
      estimate < 0 ~ "Lower complaint rate",
      TRUE ~ "No difference"
    )
  )

ggplot(
  race_results,
  aes(x = estimate, y = reorder(race, estimate))
) +
  # Zero-reference line, subdued so it reads as a guide, not a data element
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey55", linewidth = 0.4) +

  # Point + interval combined into one layer, colored by direction/magnitude
  geom_pointrange(
    aes(xmin = conf.low, xmax = conf.high, color = estimate, alpha = significant),
    size = 0.9,
    linewidth = 0.9
  ) +

  # Direct value labels above each point -- removes the need to eyeball gridlines

```

```

geom_text(
  aes(label = number(estimate, accuracy = 0.01, style_positive = "plus")),
  family = base_font,
  size = 3.4,
  color = "grey25",
  vjust = -1.4,
  fontface = "bold"
) +

# Diverging color scale: color encodes direction AND magnitude, not decoration
scale_color_gradient2(
  low = "#3B7DD8", mid = "grey60", high = "#D6604D",
  midpoint = 0, guide = "none"
) +
scale_alpha_manual(values = c(`TRUE` = 1, `FALSE` = 0.35), guide = "none") +

scale_x_continuous(
  breaks = pretty_breaks(6),
  expand = expansion(mult = c(0.12, 0.12))
) +

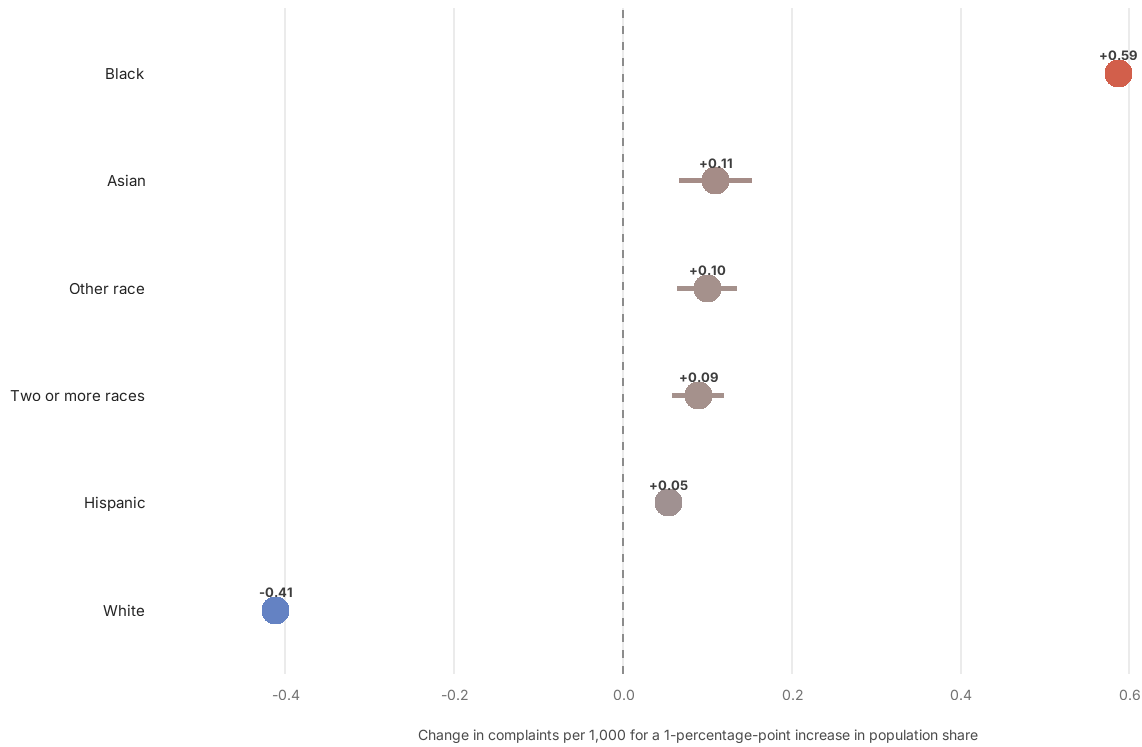
labs(
  title = "Population Composition and Credit-Reporting Complaint Rates",
  subtitle = "Each model adjusts for income, poverty, unemployment, and median age",
  x = "Change in complaints per 1,000 for a 1-percentage-point increase in population share",
  y = NULL,
  caption = "Source: [CFPB, US Census] · Bars show 95% confidence intervals"
) +

theme_minimal(base_size = 13, base_family = base_font) +
theme(
  plot.title = element_text(face = "bold", size = 16, margin = margin(b = 4)),
  plot.subtitle = element_text(color = "grey40", size = 11.5, margin = margin(b = 16)),
  plot.caption = element_text(color = "grey55", size = 8.5, hjust = 0, margin = margin(t = 12)),
  axis.title.x = element_text(color = "grey30", size = 10.5, margin = margin(t = 10)),
  axis.text.y = element_text(size = 11.5, color = "grey15"),
  axis.text.x = element_text(color = "grey45"),
  panel.grid.major.y = element_blank(),
  panel.grid.minor = element_blank(),
  panel.grid.major.x = element_line(color = "grey92", linewidth = 0.4),
  plot.margin = margin(20, 24, 16, 20)
)

```

Population Composition and Credit-Reporting Complaint Rates

Each model adjusts for income, poverty, unemployment, and median age



Source: [CFPB, US Census] - Bars show 95% confidence intervals

Race/ethnicity	Estimate	Interpretation	VIF
White	-0.412	A 1-percentage-point increase in the White population share is associated with 0.412 fewer complaints per 1,000 residents , adjusting for income, poverty, unemployment, and age.	1.459
Black	0.586	A 1-percentage-point increase in the Black population share is associated with 0.586 more complaints per 1,000 residents , adjusting for the same factors.	1.266

Race/ethnicity	Estimate	Interpretation	VIF
Asian	0.110	A 1-percentage-point increase in the Asian population share is associated with 0.110 more complaints per 1,000 residents.	1.292
Hispanic	0.053	A 1-percentage-point increase in the Hispanic population share is associated with 0.053 more complaints per 1,000 residents.	1.152
Other race	0.100	A 1-percentage-point increase is associated with 0.100 more complaints per 1,000 residents.	1.122
Two or more races	0.089	A 1-percentage-point increase is associated with 0.089 more complaints per 1,000 residents.	1.101

```
library(tidyverse)

plot_data <- complaints_by_zip %>%
  select(
    complaints_per_1000,
    median_household_income,
    poverty_percent,
    unemployment_percent,
    median_age
  ) %>%
  pivot_longer(
    cols = c(
      median_household_income,
      poverty_percent,
      unemployment_percent,
      median_age
    ),
    names_to = "predictor",
    values_to = "x"
```

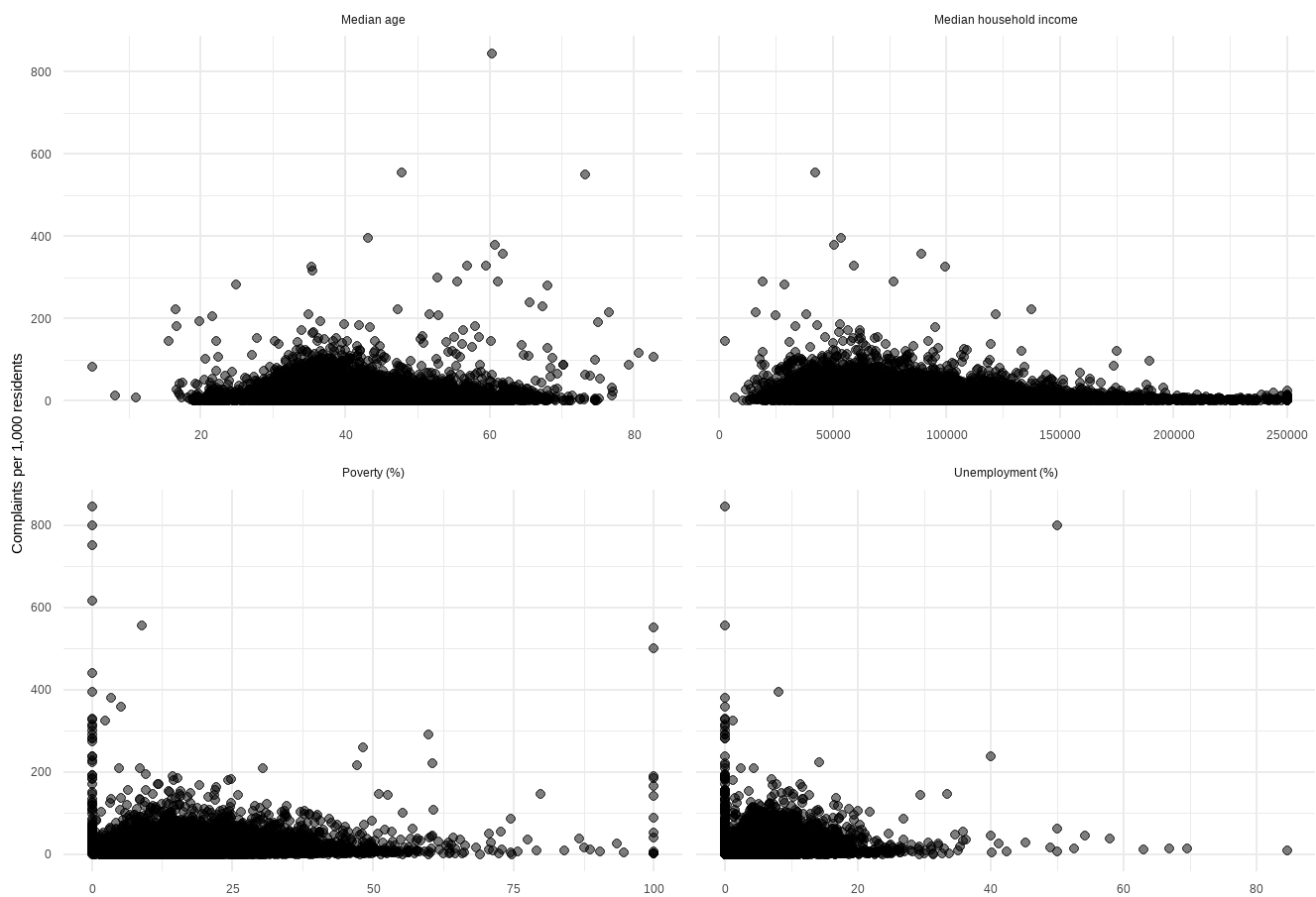
```

) %>%
mutate(
  predictor = case_when(
    predictor == "median_household_income" ~ "Median household income",
    predictor == "poverty_percent" ~ "Poverty (%)",
    predictor == "unemployment_percent" ~ "Unemployment (%)",
    predictor == "median_age" ~ "Median age"
  )
)

ggplot(
  plot_data,
  aes(x = x, y = complaints_per_1000)
) +
  geom_point(alpha = 0.5) +
  facet_wrap(~ predictor, scales = "free_x") +
  labs(
    title = "ZIP-Code Characteristics and Credit-Reporting Complaint Rates",
    x = NULL,
    y = "Complaints per 1,000 residents"
  ) +
  theme_minimal()

```

ZIP-Code Characteristics and Credit-Reporting Complaint Rates



Predictor	Estimate	Interpretation	VIF
Income (\$10,000s)	-0.407	A \$10,000 increase in median household income is associated with 0.407 fewer complaints per 1,000 residents , holding poverty, unemployment, and age constant.	1.816
Poverty (%)	0.103	A 1-percentage-point increase in poverty is associated with 0.103 more complaints per 1,000 residents , holding the other predictors constant.	2.191
Unemployment (%)	0.567	A 1-percentage-point increase in unemployment is associated with 0.567 more complaints per 1,000 residents .	1.267
Median age	-0.204	A 1-year increase in median age is associated with 0.204 fewer complaints per 1,000 residents .	1.068